# Assignment 1 - Part B

## Neural Networks with PyTorch

Kuzushiji-MNIST Classification Report

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94691 - Deep Learning | Semester 1, 2026

# 1. Introduction

This report documents the iterative development of fully-connected neural network classifiers for the Kuzushiji-MNIST (KMNIST) dataset using PyTorch. Through six hypothesis-driven experiments, each motivated by specific observations from the previous, the architecture, training strategy, and data pipeline were progressively refined from a baseline of 87.75% to a final best test accuracy of 95.96%.

# 2. Dataset Presentation

The KMNIST dataset contains 70,000 images (60k train, 10k test) of handwritten Hiragana characters, each 28x28 grayscale pixels flattened to 784-dimensional vectors. The 10 classes correspond to Hiragana rows: o, ki, su, tsu, na, ha, ma, ya, re, and wo. Pixel values were normalised to [0, 1] and labels one-hot encoded.

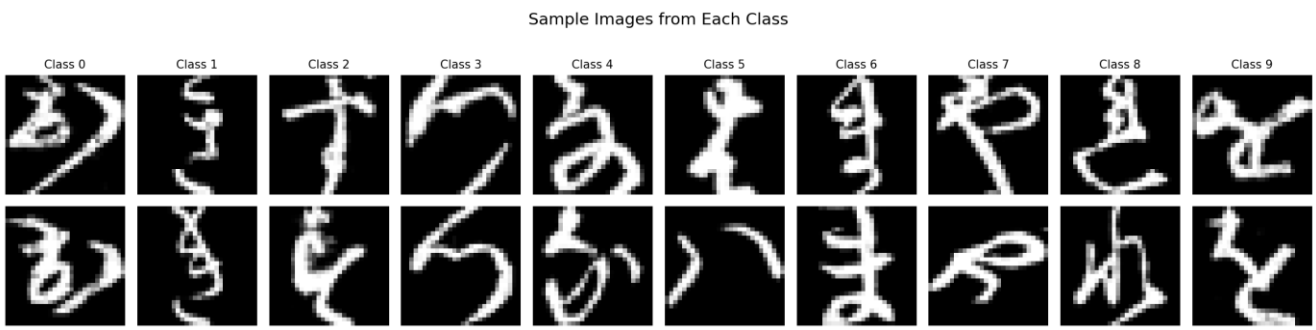


*Figure 1: Sample images from each of the 10 classes*

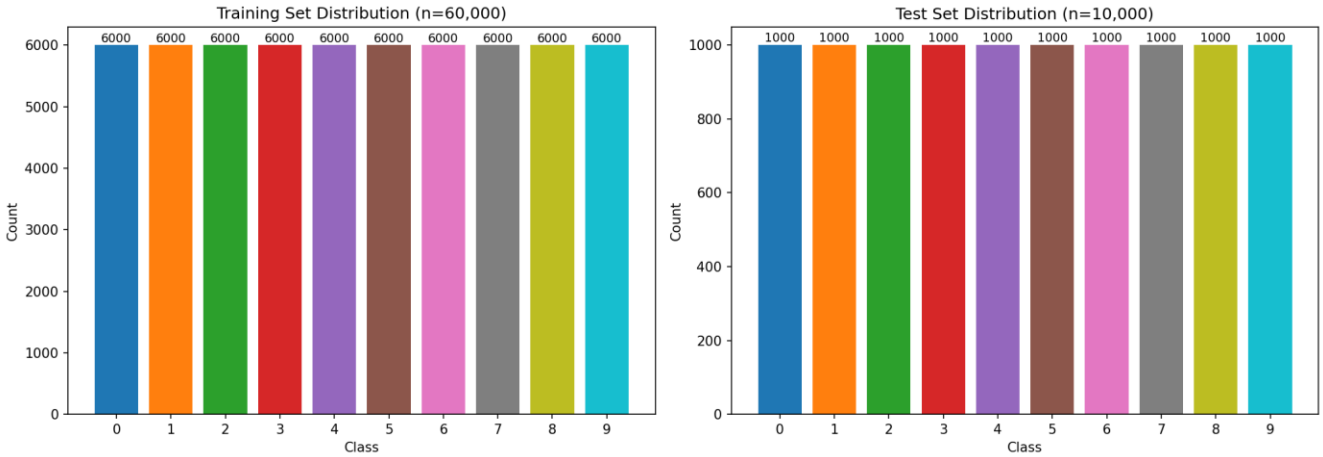


*Figure 2: Class distribution in training and test sets*

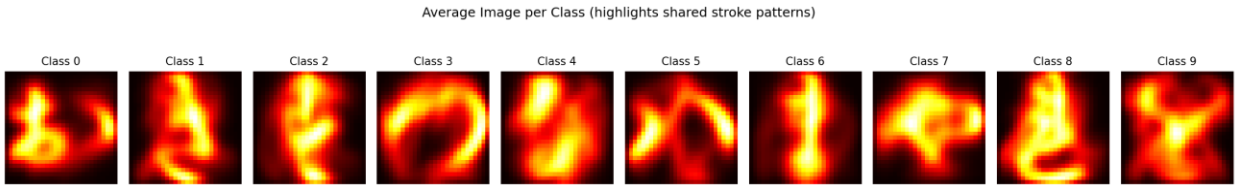


*Figure 3: Average image per class — highlights shared stroke patterns between classes*

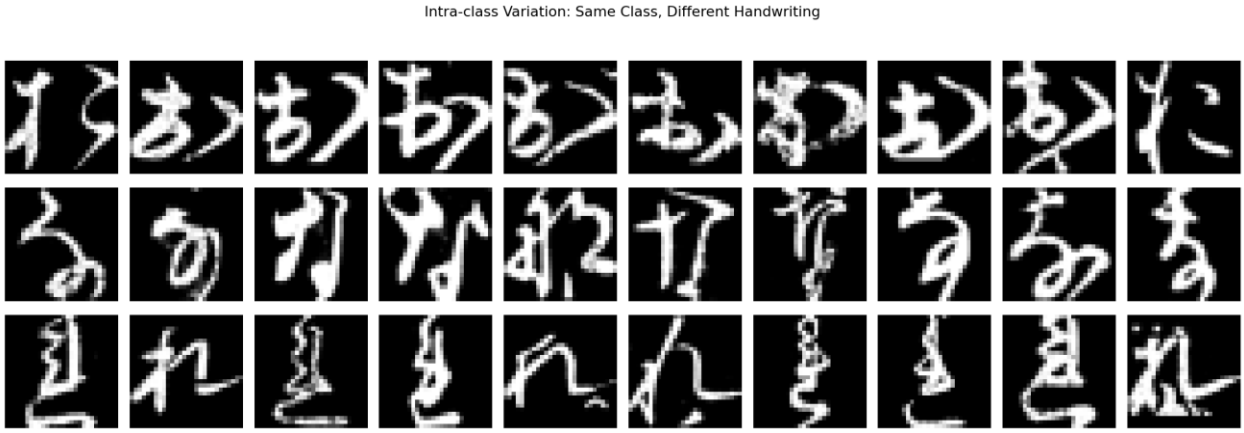


*Figure 4: Intra-class variation — same class, different handwriting styles*

The dataset is balanced with 6,000 training and 1,000 test samples per class. The average images reveal that several classes share similar stroke patterns (e.g., Classes 2 and 8), explaining why inter-class confusion is a persistent challenge. The intra-class variation shows significant diversity in handwriting style, motivating data augmentation to expose the model to more variation.

# 3. Experiments

Each experiment was designed to address a specific limitation observed in the previous one, using a null hypothesis framework.

## 3.1 Experiment 1: Shallow Baseline

- 784 → 256 → 10 (203,530 params). SGD (lr=0.01), no regularisation, 100 epochs

**Results: Train 96.87%, Test 87.75%, Gap 9.12%**

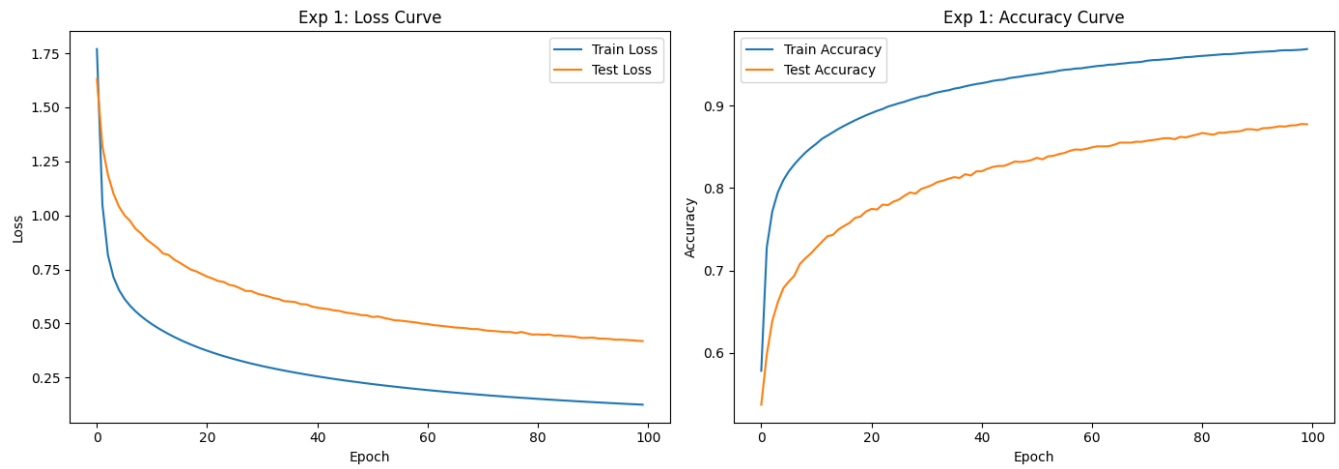


*Figure 5: Experiment 1 — Loss and accuracy curves*

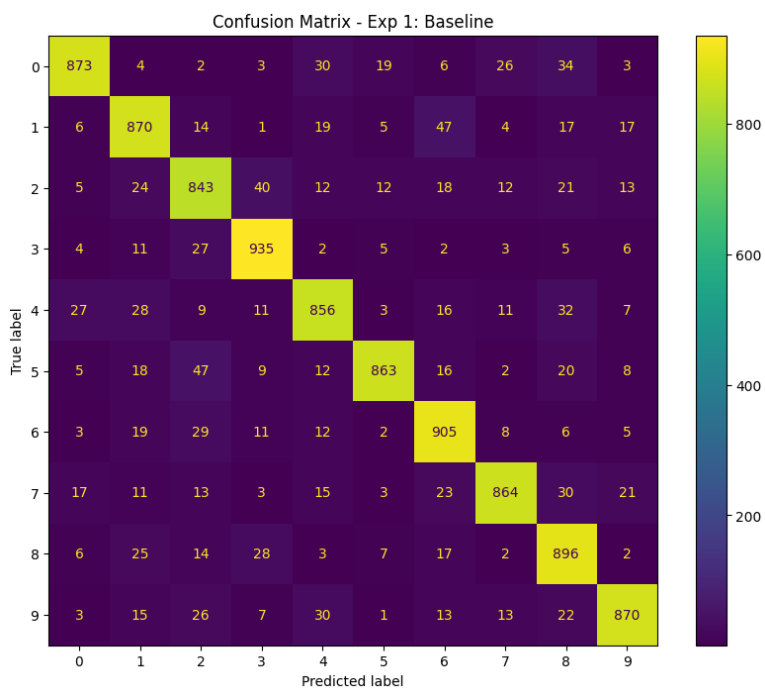


*Figure 6: Experiment 1 — Confusion matrix*

The baseline exceeds the 80% target but overfits significantly (9.12% gap). The confusion matrix reveals frequent misclassifications between visually similar classes.

## 3.2 Experiment 2: Deeper Network with Dropout

- 784 → 512 → 256 → 10 (535,818 params).
- SGD (lr=0.01),
- Dropout(0.5),
- 200 epochs

**H0: Adding depth and Dropout(0.5) will not significantly reduce the generalisation gap.**

**Results: Train 98.13%, Test 90.87%, Gap 7.26%**

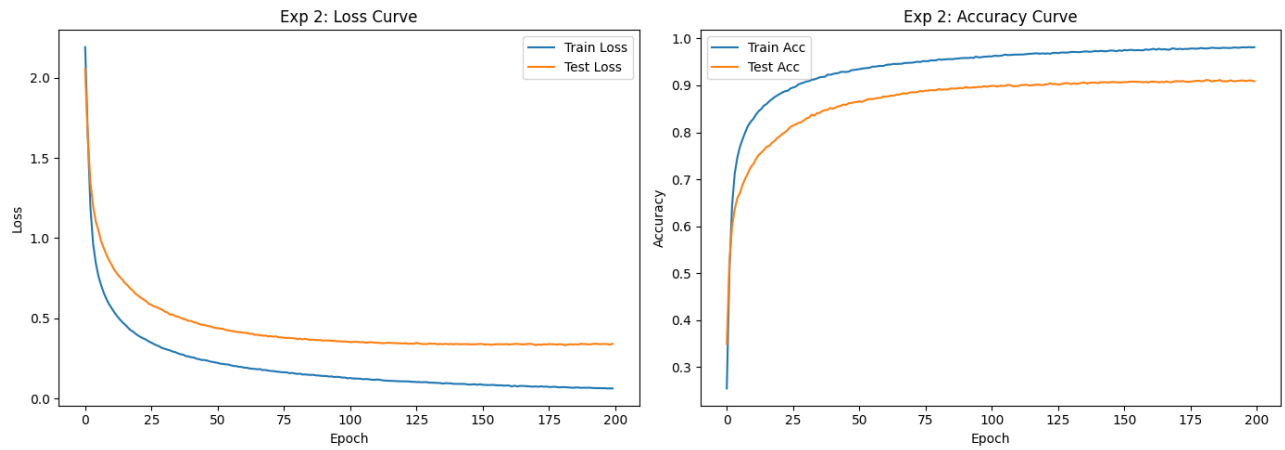


*Figure 7: Experiment 2 — Loss and accuracy curves*

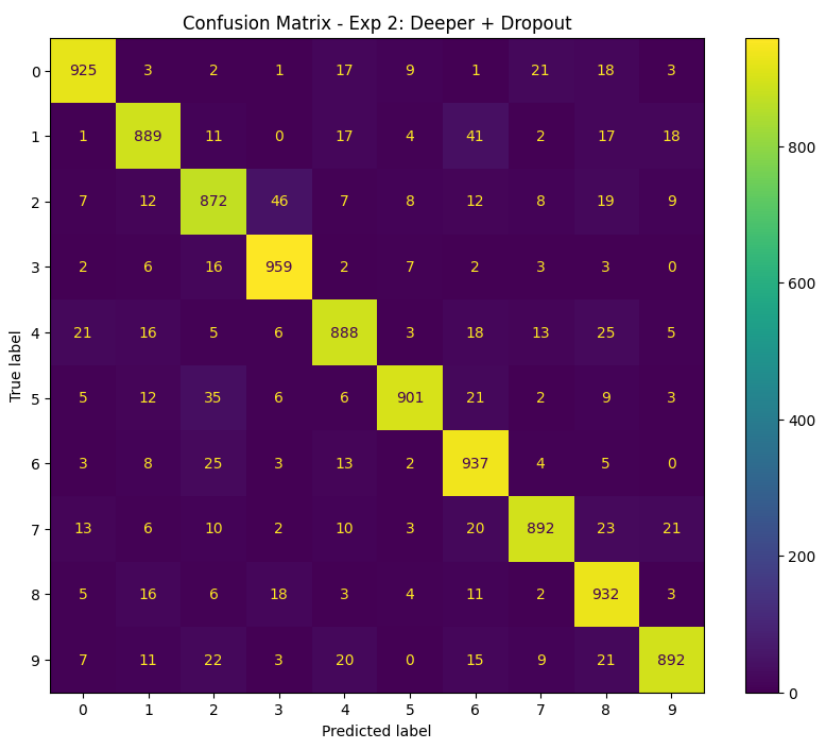


*Figure 8: Experiment 2 — Confusion matrix*

**H0 rejected**. Test accuracy improved by 3.12 points and the gap narrowed from 9.12% to 7.26%. Dropout is effective but the 0.5 rate may be constraining capacity.

## 3.3 Experiment 3: Optimised Architecture

- 784 → 1024 → 512 → 256 → 10 (1,462,538 params)
- AdamW (lr=0.001, weight_decay=1e-4),
- label smoothing 0.1,
- ReduceLROnPlateau,
- grad clip,
- 200 epochs

**H0: Wider layers with AdamW, label smoothing, and LR scheduling will not improve test accuracy beyond 90.87%.**

**Results: Train 99.97%, Test 93.39% (best 93.61%), Gap 6.58%**

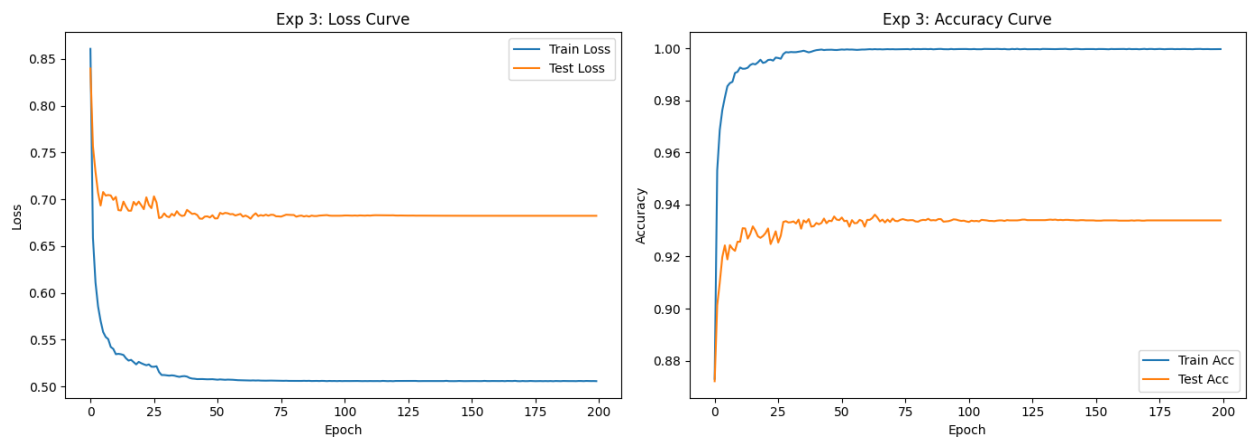


*Figure 9: Experiment 3 — Loss and accuracy curves (note: train loss floors at ~0.506 due to label smoothing)*

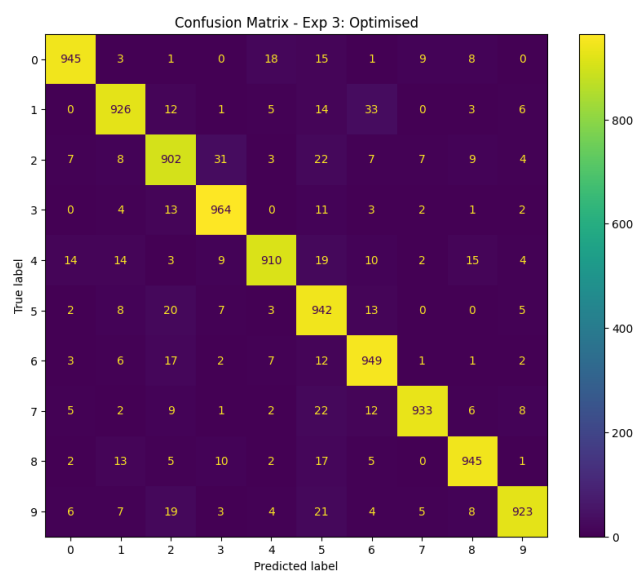


*Figure 10: Experiment 3 — Confusion matrix*

**H0 rejected**. Test accuracy reached 93.39%, but 99.97% train accuracy reveals near-complete memorisation. The parameter-to-data ratio of 24.4:1 is identified as the root cause (Zhang et al., 2017).

## 3.4 Experiment 4: 3x Augmentation

- Same architecture. 3x pre-generated augmented copies (180k images). Ratio: 8.1:1

**H0: 3x augmentation (ratio 8.1:1) will not significantly reduce memorisation.**

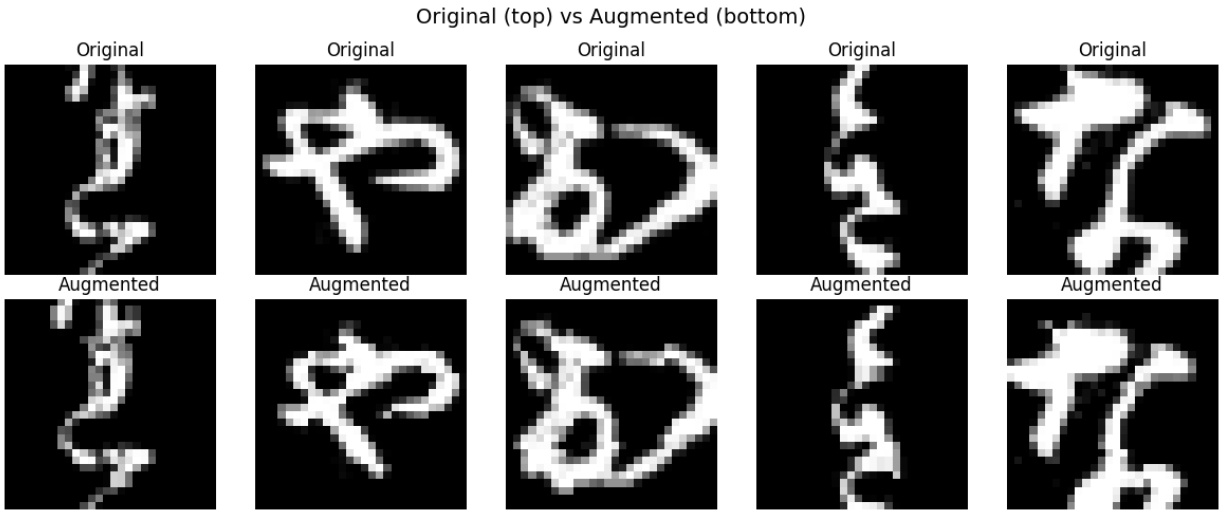


*Figure 11: Original vs augmented samples (rotation, affine transform)*

**Results: Train 99.81%, Test 93.32% (best 93.56%), Gap 6.49%**

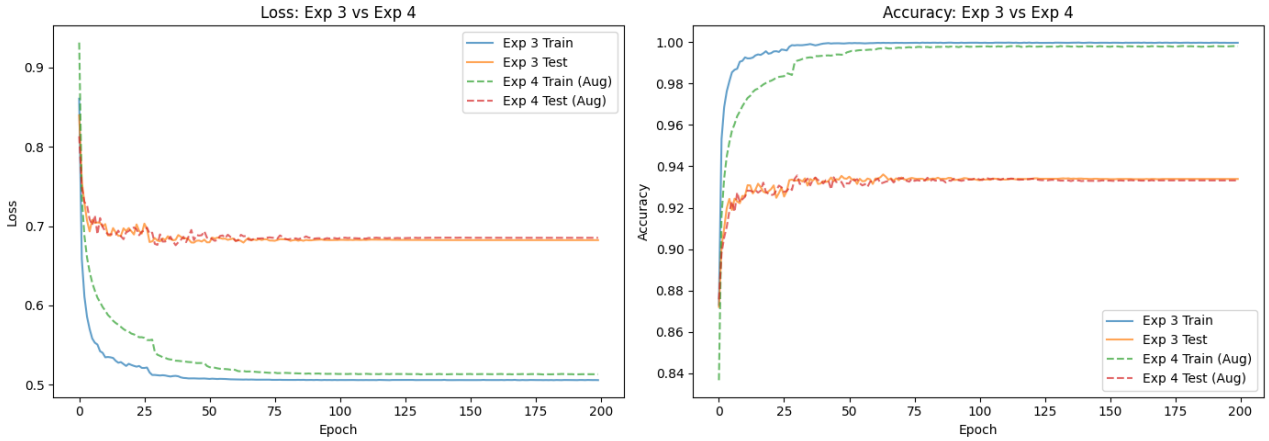


*Figure 12: Experiment 4 — Loss and accuracy curves*

However, this is a crucial result that reveals the potential of scaling the dataset size. Comparing the training loss of Experiment 4 to Experiment 3, the augmented model has clearly **more loss remaining before reaching the overfitting floor**. It has not yet fully memorised the data at the same training stage. This indicates that while 3x is insufficient, increasing the augmentation multiplier further could push the model into a regime where memorisation is no longer feasible, motivating the next experiment with a much larger dataset.

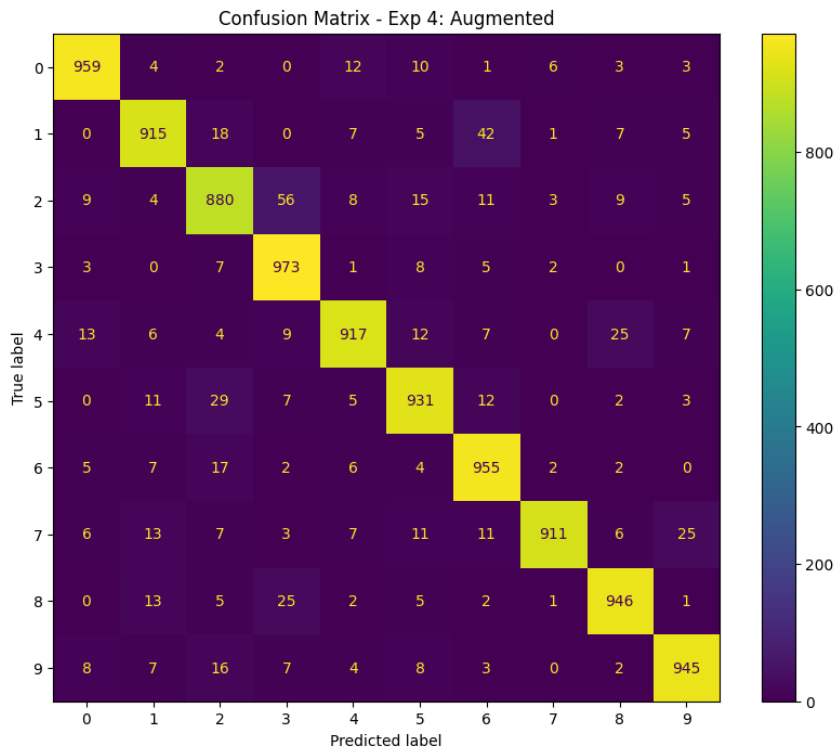


*Figure 13: Experiment 4 — Confusion matrix*

H0 not rejected. 3x augmentation (180k, ratio 8.1:1) was insufficient. The model memorised all 180k fixed copies (99.81% train). A much larger multiplier is needed.

## 3.5 Experiment 5: 12x Augmentation

- Same architecture. 12x pre-generated copies (720k images). Ratio: 2.03:1

**H0: 12x augmentation will not produce meaningful improvement beyond Experiments 3/4.**

**Results: Train 99.18%, Test 95.34%, Gap 3.84%**

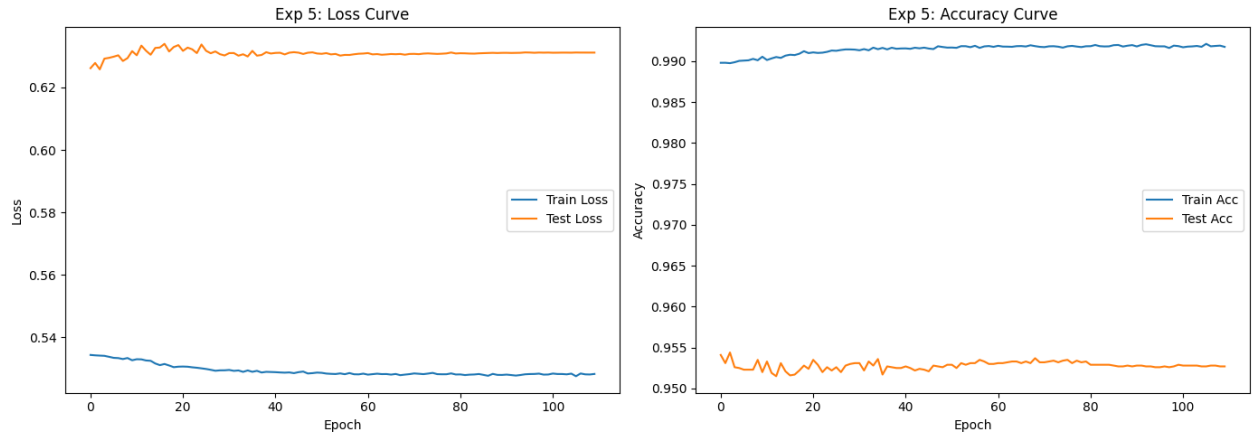


*Figure 14: Experiment 5 — Loss and accuracy curves*

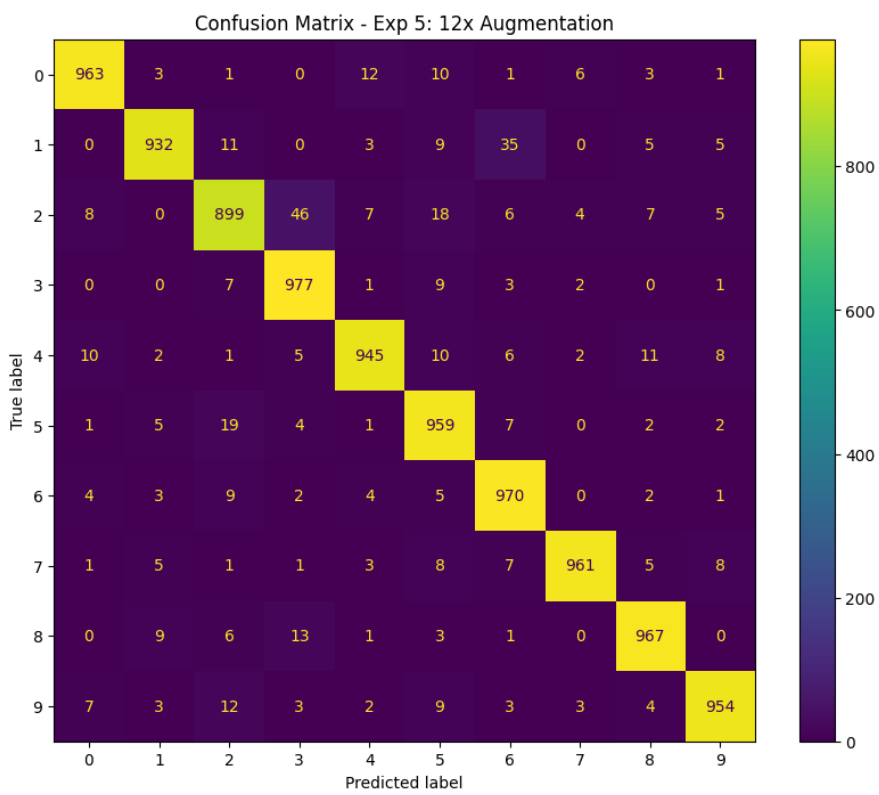


*Figure 15: Experiment 5 — Confusion matrix*

H0 rejected. Test accuracy jumped to 95.34% (+2.02 points). The gap narrowed from 6.49% to 3.84%. A critical threshold was crossed at ratio 2.03:1 where memorisation is significantly disrupted.

## 3.6 Experiment 6: 18x Augmentation (50 epochs)

- Same architecture.
- 18x pre-generated copies (1.08M images).
- Ratio: 1.35:1. 50 epochs

**H0: 18x augmentation with 50 epochs will not improve beyond Experiment 5.**

**Results: Train 98.64%, Test 95.80% (best 95.96%), Gap 2.84%**

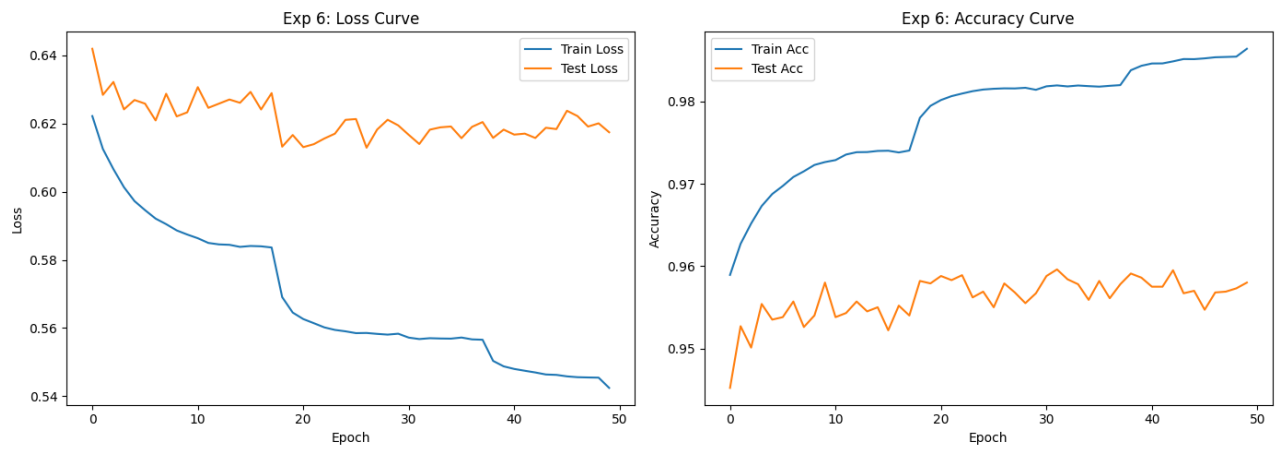


*Figure 16: Experiment 6 — Loss and accuracy curves*

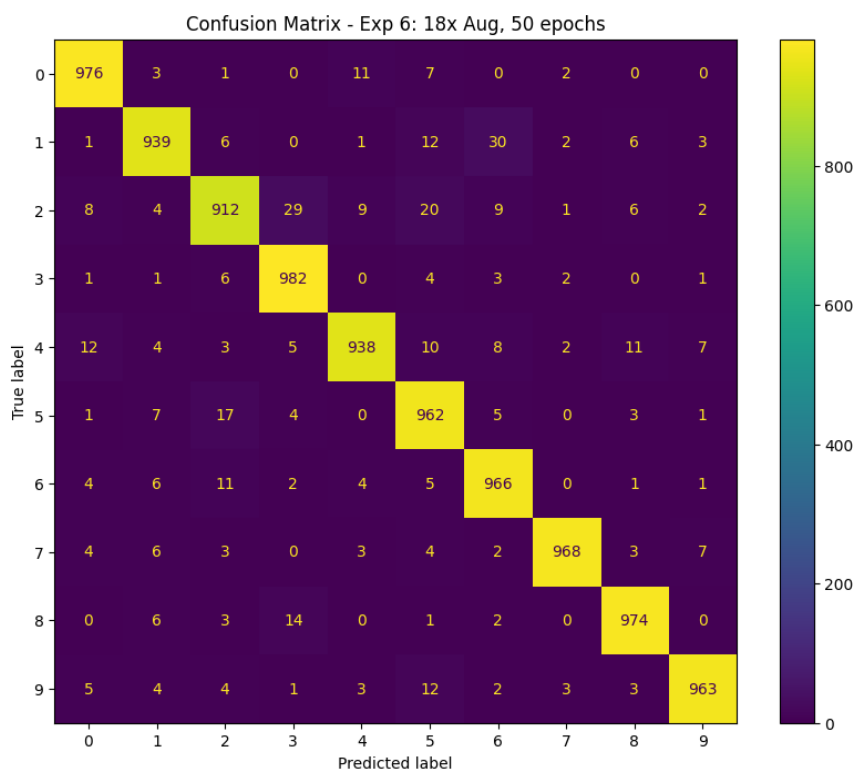


*Figure 17: Experiment 6 — Confusion matrix*

**H0 rejected**. Test accuracy improved to 95.80% (best 95.96%) with the smallest gap (2.84%). Diminishing returns from 12x→18x (+0.46 vs +2.02 points) suggest approaching the FC ceiling.

# 4. Performance Analysis

***Table 1: Summary of all experiments***

| Exp | Train | Test | Best | Gap | Params | Data | Ratio | H0 |
|---|---|---|---|---|---|---|---|---|
| 1: Baseline | 96.87 | 87.75 | - | 9.12 | 204k | 60k | 3.4:1 | - |
| 2: +Dropout | 98.13 | 90.87 | - | 7.26 | 536k | 60k | 8.9:1 | Rej. |
| 3: Optimised | 99.97 | 93.39 | 93.61 | 6.58 | 1.46M | 60k | 24.4:1 | Rej. |
| 4: +3x Aug | 99.81 | 93.32 | 93.56 | 6.49 | 1.46M | 180k | 8.1:1 | N.R. |
| 5: +12x Aug | 99.18 | 95.34 | 95.34 | 3.84 | 1.46M | 720k | 2.0:1 | Rej. |
| **6: +18x/50e** | **98.64** | **95.80** | **95.96** | **2.84** | **1.46M** | **1.08M** | **1.4:1** | **Rej.** |

*H0 column: Rej. = null hypothesis rejected, N.R. = not rejected*

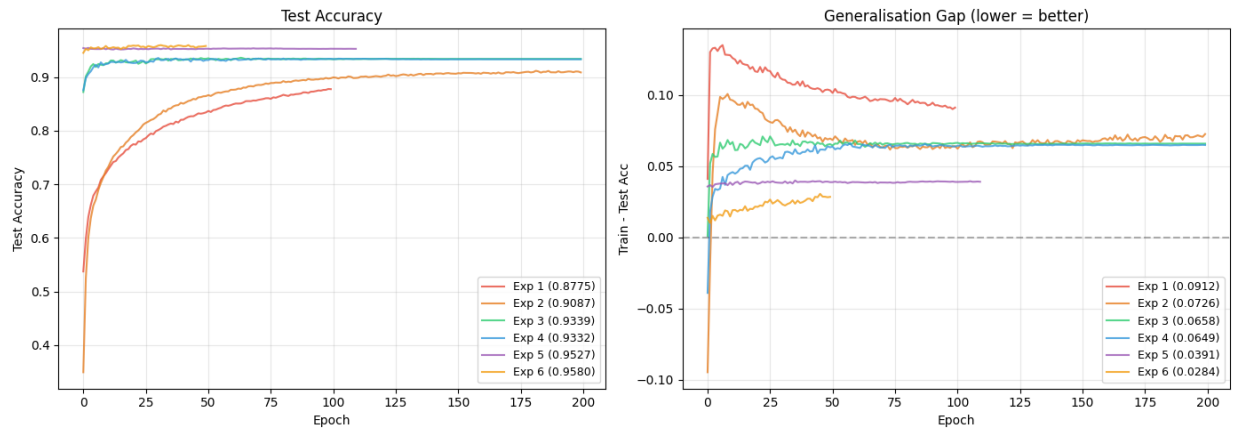


*Figure 18: Test accuracy and generalisation gap across all experiments*

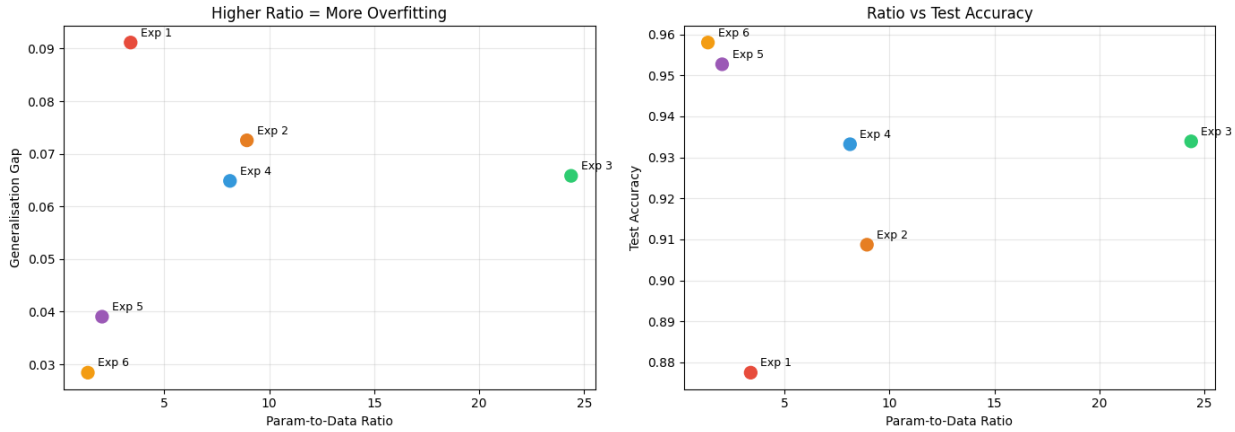


*Figure 19: Parameter-to-data ratio vs test accuracy and generalisation gap*

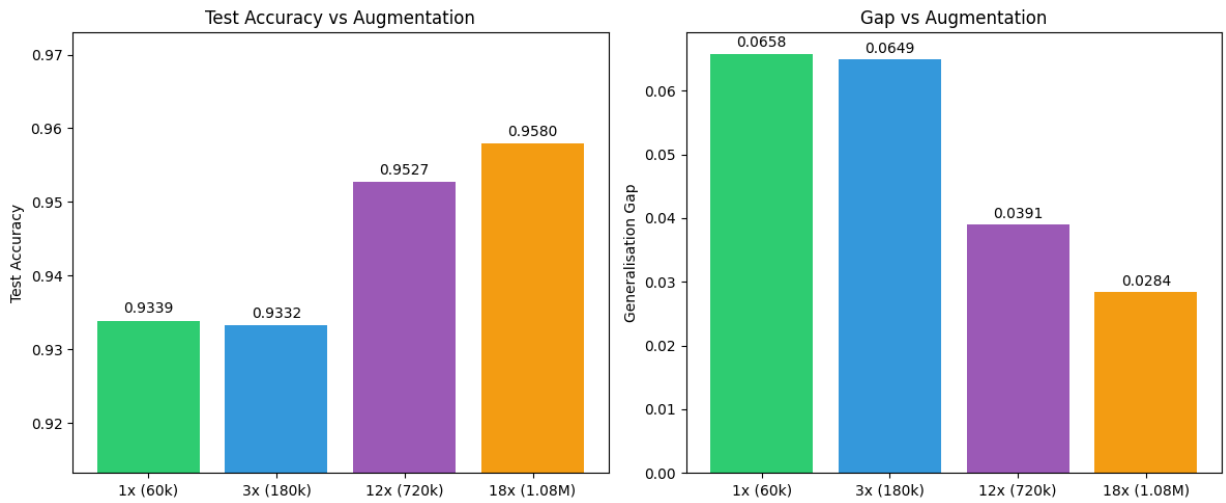


*Figure 20: Effect of augmentation multiplier on test accuracy and gap*

**Parameter-to-data ratio and memorisation:**

Zhang et al. (2017) demonstrated that overparameterised networks memorise datasets when parameters exceed training samples. Our results quantify this: at ratios above 8:1 (Experiments 3 and 4), the model achieves near-perfect training accuracy regardless of dataset size. Only at ratio 2.0:1 (Exp 5) does memorisation decrease meaningfully, with further improvement at 1.4:1 (Exp 6). Experiment 4 proves the ratio threshold matters more than absolute dataset size.

**Critical augmentation threshold:**

The most striking finding is the non-linear relationship between augmentation and performance. Going from 1x to 3x produced no improvement, while 3x to 12x yielded a 2.02-point jump. This suggests a critical threshold around ratio 2-3:1 where the model's memorisation capacity is exceeded. Arpit et al. (2017) showed networks learn generalisable patterns before memorising; once augmentation exceeds this threshold, useful learning is preserved while memorisation is disrupted.

**Diminishing returns:**

The improvement from 12x to 18x (+0.46 points) is much smaller than 3x to 12x (+2.02 points). With pre-generated augmentation, the practical ceiling for FC networks on KMNIST appears to be approximately 96%.

## 5. Limitations and Recommendations

**Loss of spatial information:**

Flattening images discards spatial relationships. Hiragana strokes encode meaning in position and curvature, fundamentally limiting FC accuracy.

**Pre-generated vs on-the-fly augmentation:**

Pre-generated copies are fixed and can eventually be memorised. On-the-fly augmentation provides infinite diversity but was prohibitively slow due to CPU-bound transforms. GPU-native augmentation (e.g., Kornia) could combine speed with infinite diversity.

**Recommendations:**

- **On-the-fly augmentation:** GPU-native on-the-fly augmentation (Kornia) for infinite diversity without CPU bottleneck.
- **Model ensembling:** Training multiple models with different seeds and averaging predictions.
- **Convolutional networks:** CNNs preserve spatial structure and achieve >95% on KMNIST benchmarks.

## 6. Conclusion

Through six hypothesis-driven experiments, a fully-connected network achieving 95.96% best test accuracy was developed on KMNIST with a generalisation gap of 2.84%. The key findings are: (1) dropout and depth improve baseline performance (H0 rejected); (2) wider architecture with modern optimisation pushes to 93%+ but causes memorisation (H0 rejected); (3) insufficient augmentation fails despite tripling data (H0 not rejected); (4) a critical threshold exists around ratio 2:1 where memorisation is disrupted (H0 rejected); and (5) further augmentation yields diminishing but measurable returns (H0 rejected). The parameter-to-data ratio governs memorisation in overparameterised FC networks, confirming Zhang et al. (2017) and Arpit et al. (2017).

## References

Zhang, C., Bengio, S., Hardt, M., Recht, B., & Vinyals, O. (2017). Understanding deep learning requires rethinking generalization. ICLR 2017. arXiv:1611.03530

Arpit, D., Jastrzebski, S., Ballas, N., et al. (2017). A closer look at memorization in deep networks. ICML 2017. arXiv:1706.05394

Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. Journal of Big Data, 6(1), 1-48.